**Qualcomm**

Position: Digital Design Engineer

Location: San Diego, CA

**Phone Interview:**

20 min discussion on projects I did

1) What is the width (number bits) of result if five 8-bit two's complement numbers are adders?

2) What are the possible ways can we send an 8-bit word from a clock-A to clock-B?

3) For a given FIFO which can write 80 words in 100 clocks and read 8 words in 10 clocks, what is the worst case stack depth needed for an input burst of 160 words?

**Onsite Interview:**

1) Draw the circuit for frequency divider-by -3 with 33% duty cycle. Write Verilog code for this ckt.
Write the Verilog code for the frequency divider using a 2-bit counter.

2) Write the code for a register with synchronous and asynchronous reset.

3) How do you synchronize the signals between two clock domains?

4) We have 3 pipeline stage registers FF-1, FF-2 and FF-3. The input to FF-1 is 'd' , and the output of FF-1 'q1' is connected to the input of FF-2, similarly, the output of FF-2 'q2' is connected to the input of FF-3, whose output is 'q3'. Does the below Verilog code synthesize to the pipeline stages as described above? If yes, how? If not, how?

```
always@(posedge clk)
begin
  q1 = d;
  q2 = q1;
  q3 = q2;
end
```

5) Change the above implementation to non-blocking assignments, what is the synthesized circuit?

6) Change the order of the assignments in above code as below, what is the synthesized circuit?

```
always@(posedge clk)
begin
  q3 = q2;
  q2 = q1;
  q1 = d;
end
```

7) Does order of assignment matters in non-blocking assignment (in ques-5)?

8) We have a 16-bit data input feeding to a register and passing this value to 95 other registers for every clock cycle (i.e. R0, R1…R95). The block is similar to 96 pipeline stages. The output of each of the registers R0, R1…R95 is multiplied with a 16-bit scalar term (say 'k') and given as an input to an Adder. So, we have 96 multipliers and each multiplier output is given as input to an Adder, which adds all the inputs and outputs a value 'N'. Input data is operating at 25 MHz.
What is the bit length of output 'N'?

What are the minimum number of Adders we need to implement this, with internal logic operating at frequencies 25 MHz, 50 MHz, 100 MHz and 200 MHz?
What is the memory architecture we can follow in order to have high through-put?
9) In C-language, what is the difference between symbols & and *?
10) Write a Verilog code for Mux, why the output is declared as 'reg' ?
11) We have four registers, all operating in same clock frequency and all samples input value concurrently (say A[0], A[1], A[2] & A[3]). What are the modifications we need to do to the above circuit in such a way that we can sample the values for both the clock edges?
12) What are the different types of power dissipations in a CMOS transistor?
13) In an IDEAL CMOS inverter, how do we size PMOS and NMOS and why?
14) We have 25 Red and 25 Blue beads and you need to arrange them in two bowls in such a way that I will randomly choose one bowl and select one bead. What is way we need to arrange the beads in both the bowls, such that I have maximum probability that the bead I select will be a Red one.
15) What is metastability and how it is caused?
16) What are the ways I can reduce metastability?
17) What are the ways I synchronize the signal between different clock domains, in such a way that I collect the data (without any loss) at receiving end?
18) I have a signal coming out of an ASIC board (operates at 10 MHz), which is feed to an FPGA board (which operates at 1 MHz). Now, construct a circuit in FPGA board in such a way that I will receive the pulse from ASIC board (with period of 100 ns = 100 MHz) in FPGA board and I need to send back another pulse (with period of 1 us = 1 MHz)?
19) If we have 32 words (each 8-bit length), what are the ways we have to store these values?
What are the parameters we consider to choose the best implementation?
(The discussion went on to the implementation in register bank, SRAM and DRAM. The implementation criteria, area and throughput comparison).
20) I have a RAM which can write-in and read-out the values simultaneously using the signals "w_en" and "r_en". I have an "address", data-input "Din" and data-output "Dout". Write a verilog code for this RAM.
21) I have a function y = a*b in which 'a' and 'b' are complex numbers. How many multipliers I need, in order to implement this function in such a way that I get the output for every clock cycle?
22) The output of a positive edge triggered flipflop FF-1 is connected to a negative flipflop FF-2 input, via a combinational logic, whose propagation delay is $T_G$. FF-1 and FF-2 are clocked with Negative skew (in FF-2 clock path). What is the maximum frequency that this circuit operates on?
23) Describe the working of an inverter.
24) In a ring oscillator, what happens to the output signal if I connect a capacitance to the input of last stage?
25) Design a FIFO which has its writing end writes 80 words in 100 clock cycles and reading side reads 8 words in 10 cycles. In 100 cycles (of writing side), I have 20 idle cycles which can come in any order. I have two bursts of date each with 80 words. What is the worst case of arranging the write and idle cycles for these two bursts? What is the worst case FIFO depth?